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Method and apparatus for securing concealable body support

Abstract

A walk-in bathing environment is used by people who may have trouble getting in and out of standard bathtubs or showers. Rotatable grab bars are positioned in and around the bathing environments to aid in entering the environments, exiting the environments, sitting, standing, and otherwise moving relative to the bathing environment. The rotatable grab bars may be placed on the walls, top, external, or generally on and around the bathing environment such that a user can grab the one or more rotatable grab bars in order to provide support for moving relative to the bathing environment. The rotatable grab bars are connected to the environment such that they can be moved between a use and a storage configuration, such that the rotatable grab bars are extending from a portion of the housing only when desired by the user.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to the use of collapsible, rotatable, hideable, and/or concealable grab bars. The grab bars may be used, for example, with walk-in bathtubs or showers, in combination with seating or furniture, or even in at least boating, airline, fitness, and home appliance industries.

BACKGROUND

(2) It can be difficult for many to get in and out of a regular height bathtub, and even harder for the elderly and those with disabilities. For some people, it is simply impossible. Wet slippery surfaces and high walls can and do cause accidents and injuries. In fact, statistics show that most accidents in the home occur in the bathroom.

(3) Walk-in bathtubs and showers provide environments that attempt to alleviate the issues of getting in and out of regular bathtubs. A user can simply step over the low threshold, onto the slip resistant floor of the tub, shut the door, sit down on the comfortable chair height seat, and fill the bathtub with calming water.

(4) Furthermore, to aid in entering, exiting, or using the bathtub or shower, grab bars may be included and strategically placed in or around the bathtub or shower. These grab bars, which can be one or a plurality of bars, can be angled or otherwise oriented to provide the user with ease of sitting or standing with the least amount of effort. This is also true for entering and exiting of the tub, as the grab bars can provide a surface for stabilization.

(5) However, as the bars extend from a surface of the bathtub or shower, they can create obstacles or obstructions when not in use. The bars extend away from the bathtub or shower surface and can injure a user if a fall or slip were to occur.

(6) Therefore, there is a need in the art for a grab bar that can be selectively provided for use when needed, but that can be moved to a non-use configuration in a user-friendly manner in which the grab bar is out of a user's way.

SUMMARY

(7) The following objects, features, advantages, aspects, and/or embodiments are not exhaustive and do not limit the overall disclosure. No single embodiment need provide each and every object, feature, or advantage. Any of the objects, features, advantages, aspects, and/or embodiments disclosed herein can be integrated with one another, either in full or in part.

(8) It is a primary object, feature, or advantage of the present invention to improve on or overcome the deficiencies in the art.

(9) It is still yet a further object, feature, or advantage of the present invention to provide a grab bar that can be locked into place during use or when the grab bar is in the non-use configuration, thereby preventing injury to a user.

(10) It is still yet a further object, feature, or advantage of the present invention to provide a grab bar that is cost effective.

(11) It is still yet a further object, feature, or advantage of the present invention to provide a grab bar that is reliable and durable and has a long usable life.

(12) It is still yet a further object, feature, or advantage of the present invention to provide a grab bar that is easily manufactured, assembled, installed, disassembled, and uninstalled.

(13) It is still yet a further object, feature, or advantage of the present invention to provide an apparatus that is aesthetically pleasing.

(14) It is still yet a further object, feature, or advantage of the present invention to incorporate the grab bar into a system which includes a walk-in bathtub and/or a shower.

(15) It is still yet a further object, feature, or advantage of the present invention to provide methods of using, manufacturing, and installing a grab bar accomplishing some or all of the previously stated objectives.

(16) The present disclosure is directed generally towards one or more grab bars for use with walk-in bathtubs and/or showers. The bathtubs and/or showers may be referred to as a bathing environment of the walk-in sort. A walk-in bathing environment is used by people who may have trouble getting in and out of standard bathtubs or showers. Therefore, the bathing environments of the walk-in sort may include a door or other threshold in which a user does not need to step over in order to enter the tub or other environment.

(17) Grab bars are positioned in and around the bathing environments to aid in entering the environments, exiting the environments, sitting, standing, and otherwise moving relative to the bathing environment. The grab bars may be placed on the walls, top, external, or generally on and around the bathing environment such that a user can grab the one or more grab bars in order to provide support for moving relative to the bathing environment.

(18) The present disclosure contemplates the use of a collapsible type grab bar or a plurality of collapsible type grab bars for use with a walk-in bathing environment. The collapsible grab bars may take many forms, including but not limited to, sliding grab bars, folding grab bars, or the like. Sliding grab bars are movable to extend in and out of a grab bar housing in a manner such that the grab bars move between a useable and non-use configuration. When in a use configuration, the grab bars extend generally away from the housing such that a user is able to grab a portion of the bar to provide support, such as for standing up from a seat of the bathing environment or for entering or exiting through a door of the bathing environment. In a non-use configuration, the bar may be slid such that a majority or substantial portion of the bar is at or below the top threshold of the grab bar housing such that the grab bar is hidden away. In such a configuration, the grab bar will not extend to an area within or outside of the walk-in bathing environment to aid in mitigating the amount of surfaces for getting in the way of a user of the bathing environment.

(19) Other varieties of grab bars include folding type grab bars wherein the grab bar is foldable

between a use and a non-use configuration. The grab bar can be rotated about the grab bar housing generally between a 90° angle of movement to place the grab bar in a use or non-use configuration. For example, in a non-use configuration, the grab bar can be housed substantially within a recess of the grab bar housing such that it does not extend outwardly therefrom. However, in a use configuration, the grab bar can be rotated and locked in place such that it is rotated approximately 90° from the non-use configuration to allow a use to be able to grab a bar and use said bar in moving relative to the bathing environment. For example, when a user is done with their bathing and wishes to exit the bathing environment, they may fold down one or more grab bars such that the user can pull himself or herself up and provide a more stabilized surface for exiting the bathing environment.

(20) According to some aspects of the present disclosure, a grab bar assembly including a housing which includes both of a recessed portion and a flange portion, and the grab bar assembly further includes a central pivot pin positioned in the recessed portion. Further, the grab bar assembly includes a handle comprising a grabbing portion, a handle base, a first lock hole, and a second lock hole, wherein the handle is rotatable around the central pivot pin between a use configuration and a storage configuration. Further still, the grab bar assembly includes a button on the flange portion extending into the recessed portion and positioned on a first side of the handle, the button including a pair of button slides. Moreover the grab bar assembly includes a bolt extending through the button, through the recessed portion, and into either of the first lock hole or the second lock hole; wherein the bolt includes a pair of bolt slides in sliding communication with the pair of button slides so as to be capable of converting a first translational motion into a second translational motion orthogonal to the first translational motion when the pair of button slides interface with the pair of bolt slides.

(21) According to some additional aspects of the present disclosure, the central pivot pin passes through the recessed portion, the handle base, and the recessed portion again.

(22) According to some additional aspects of the present disclosure, an end of the central pivot pin is male-threaded and a central pivot pin aperture receiving the central pivot pin is female-threaded.

(23) According to some additional aspects of the present disclosure, the grab bar assembly further includes a spring pin to bias the bolt towards the handle, wherein the spring pin extends through the recessed portion and into the bolt.

(24) According to some additional aspects of the present disclosure, the handle includes a catch tab for preventing the rotatable body support from rotating more than 180°.

(25) According to some additional aspects of the present disclosure, the housing includes a first catch indent positioned at a second side of the handle, wherein the housing includes a second catch indent positioned at the first side of the handle, and wherein when the handle rotates from the use configuration to the storage configuration a first side of the catch tab interfaces with the first catch indent and a second side of the catch tab interfaces with the second catch indent.

(26) According to some additional aspects of the present disclosure, the grab bar assembly further includes a canyon in the recessed portion to allow for a full 180° range of motion for the handle.

(27) According to some additional aspects of the present disclosure, the button further includes a button top substantially flush with the flange, a button bottom opposite the button top, and a pair of button columns connecting the button bottom and the button top.

(28) According to some additional aspects of the present disclosure, the grab bar assembly further includes a fitted button aperture to receive and stabilize the button bottom.

(29) According to some additional aspects of the present disclosure, the bolt includes a bolt head and a bolt tail opposite the bolt head for inserting into the handle, wherein the bolt head is wider than the bolt tail.

(30) According to some additional aspects of the present disclosure, the bolt tail extends through the pair of button columns, the button top, and the button bottom, but the bolt head cannot extend through the pair of button columns, the button top, and the button bottom.

- (31) According to some additional aspects of the present disclosure, the button bottom prevents the button top from extending further than the flange.
- (32) According to some additional aspects of the present disclosure, the pair of button slides extend from the pair of button columns.
- (33) According to some additional aspects of the present disclosure, the first lock hole and the second lock hole are not aligned across the handle base.
- (34) According to some additional aspects of the present disclosure, the grab bar assembly further includes mounting apertures positioned at distal ends of the flange.
- (35) According to some additional aspects of the present disclosure, the spring pin extends through a button aperture of the grab bar assembly.
- (36) According to some additional aspects of the present disclosure, the handle base is substantially flush with the flange when in the storage position.
- (37) According to some additional aspects of the present disclosure, the housing provides stability for each of the handle, the button, the bolt, the spring pin, and the central pivot pin.
- (38) According to some other aspects of the present disclosure, a grab bar assembly built into a surface comprising: a grab bar housing including a recessed portion and a flange portion; a central pivot pin positioned in the recessed portion; a handle comprising a grabbing portion, a handle base, a first lock hole, and a second lock hole, wherein the handle is rotatable around the central pivot pin between a use configuration and a storage configuration; a button on the flange portion extending into the recessed portion and positioned on a first side of the handle, the button including a button slide; and a bolt extending into either of the first lock hole or the second lock hole, the bolt including a bolt slide for interfacing with the button slide; wherein in the storage configuration the grabbing portion is concealed within a wall of a structure, said structure being a building, a seat, or an appliance; wherein the grabbing portion is at or below the surface when in the storage configuration; and wherein the grabbing portion extends outward at least partially from the surface in the use configuration.
- (39) According to some other aspects of the present disclosure, a method of concealing and revealing a handle, comprising: housing the handle comprising a grabbing portion, a handle base, and a central pivot pin aperture with a central pivot pin extending therethrough within a housing having a recessed portion and a flange, said recessed portion both receiving and supporting the central pivot pin and said flange extending outwardly of the recessed portion and having a thickness less than a depth of the recessed portion; allowing the rotatable body support to rotate approximately 180° around the central pivot pin between a use configuration and a storage configuration, wherein in the storage configuration the housing conceals the grabbing portion within a surface of a building, a seat, or an appliance; and pressing a button positioned on the flange to lock or unlock the handle to allow for rotation from the use configuration to the storage configuration; wherein the grab bar housing is mounted to the building, the seat, or the appliance such that at least a portion of the grab bar housing is flush with the surface.
- (40) The preceding provides a list of aspects or embodiments disclosed herein and does not limit the overall disclosure. It is contemplated that any of the embodiments disclosed herein can be combined with other embodiments, either in full or partially, as would be understood from reading the disclosure. These and/or other objects, features, advantages, aspects, and/or embodiments will become apparent to those skilled in the art after reviewing the following brief and detailed descriptions of the drawings. The present disclosure encompasses (a) combinations of disclosed aspects and/or embodiments and/or (b) reasonable modifications not shown or described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Several embodiments in which the present disclosure can be practiced are illustrated and described in detail, wherein like reference characters represent like components throughout the several views. The drawings are presented for exemplary purposes and may not be to scale unless otherwise indicated.
- (2) FIG. 1 is a perspective view of a walk-in bathtub.
- (3) FIG. 2 is a top plan view of a walk-in bathtub.
- (4) FIG. 3A is a perspective view of a grab bar in a non-use configuration according to some aspects of the invention.
- (5) FIG. 3B is an additional perspective view of a grab bar in a non-use configuration according to some aspects of the invention.
- (6) FIG. 4 is a perspective view of a handle that is used with the grab bar of FIGS. 3A-3B.
- (7) FIG. 5 is a front view of the handle of FIG. 4.
- (8) FIG. 6A is a top plan view of the handle of FIG. 4.
- (9) FIG. 6B is a bottom plan view of the handle of FIG. 4.
- (10) FIG. 7A is a first elevation view of the handle of FIG. 4.
- (11) FIG. 7B is a second elevation view of the handle of FIG. 4.
- (12) FIG. 8 is a perspective view of a button that is used with the grab bar of FIGS. 3A-3B.
- (13) FIG. 9A is a first elevation view of the button of FIG. 8.
- (14) FIG. 9B is a second elevation view of the button of FIG. 8.
- (15) FIG. 10A is a top plan view of the button of FIG. 8.
- (16) FIG. 10B is a bottom plan view of the button of FIG. 8.
- (17) FIG. 11A is a third elevation view of the button of FIG. 8.
- (18) FIG. 11B is a fourth elevation view of the button of FIG. 8.
- (19) FIG. 12 is a perspective view of a spring pin that is used with the grab bar of FIGS. 3A-3B.
- (20) FIG. 13 is a perspective view of a central pivot pin that is used with the grab bar of FIGS. 3A-3B.
- (21) FIG. 14 is a perspective view of a bolt that is used with the grab bar of FIGS. 3A-3B.
- (22) FIG. 15A is a first elevation view of the bolt of FIG. 14.
- (23) FIG. 15B is a second elevation view of the bolt of FIG. 14.
- (24) FIG. 16A is a third elevation view of the bolt of FIG. 14.
- (25) FIG. 16B is a fourth elevation view of the bolt of FIG. 14.
- (26) FIG. 17A is a top plan view of the bolt of FIG. 14.
- (27) FIG. 17B is a bottom plan view of the bolt of FIG. 14.
- (28) FIG. 18A is a first perspective view of a housing that is used with the grab bar of FIGS. 3A-3B.
- (29) FIG. 18B is a second perspective view of a housing that is used with the grab bar of FIGS. 3A-3B.
- (30) FIG. 19A is a first elevation view of the housing of FIGS. 18A-18B.
- (31) FIG. 19B is a second elevation view of the housing of FIGS. 18A-18B.
- (32) FIG. 20 is a third elevation view of the housing of FIGS. 18A-18B.
- (33) FIG. 21A is a bottom plan view of the housing of FIGS. 18A-18B.
- (34) FIG. 21B is a top plan view of the housing of FIGS. 18A-18B.
- (35) FIG. 22A is another perspective view of the housing of FIGS. 18A-18B.
- (36) FIG. 22B is a cross-sectional view through a center of the housing shown in FIG. 22A.
- (37) FIG. 23A is a first elevation view of the grab bar.
- (38) FIG. 23B is a second elevation view the grab bar.
- (39) FIG. 23C is a cross-sectional view through a center of the grab bar shown in FIG. 23B.
- (40) FIG. 24A bottom elevation view of the grab bar.

- (41) FIG. 24B is a top elevation view of the grab bar.
- (42) FIG. 25A is a third elevation view of the grab bar.
- (43) FIG. 25B is a fourth elevation view of the grab bar.
- (44) FIG. 26A is a zoomed-in perspective view of the grab bar, emphasizing a locking mechanism which will be used as a reference figure to show varying cross-sectional views in FIGS. 26B and 26C.
- (45) FIG. 26B is a first cross-sectional view of the grab bar of FIG. 26A at a cross-section of about 15% of the way through the grab bar to emphasize components of the locking mechanism.
- (46) FIG. 26C is a second cross-sectional view of the grab bar of FIG. 26A at a cross-section of about 25% of the way through the grab bar to emphasize components of the locking mechanism.
- (47) An artisan of ordinary skill in the art need not view, within isolated figure(s), the near infinite distinct combinations of features described in the following detailed description to facilitate an understanding of the present disclosure.

DETAILED DESCRIPTION

- (48) The following definitions and introductory matters are provided to facilitate an understanding of the present invention. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the present invention pertain.
- (49) The terms “a,” “an,” and “the” include plural referents unless context clearly indicates otherwise. Similarly, the word “of” is intended to include “and” unless context clearly indicate otherwise. The word “or” means any one member of a particular list and also includes any combination of members of that list.
- (50) The terms “invention” or “present invention” as used herein are not intended to refer to any single embodiment of the particular invention but encompass all possible embodiments as described in the specification and the claims.
- (51) Terms such as first, second, vertical, horizontal, top, bottom, upper, lower, front, rear, end, sides, concave, convex, and the like, are referenced according to the views presented. These terms are used only for purposes of description and are not limiting unless these terms are expressly included in the claims. Orientation of an object or a combination of objects may change without departing from the scope of the invention.
- (52) The following embodiments are described in sufficient detail to enable those skilled in the art to practice the invention however other embodiments may be utilized. Mechanical, procedural, and other changes may be made without departing from the spirit and scope of the invention. Accordingly, the scope of the invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.
- (53) FIG. 1 shows a perspective view of a bathing environment **10**, which is shown to be a walk-in type bathtub. While a bathtub is shown as the bathing environment **10**, it should be appreciated that other types of bathing environments, including other types of bathtubs, showers, saunas, or other environments are included to be used with the aspects of the invention. The walk-in bathtub **10** includes a housing **12**. The housing **12** comprises a fiberglass shell that is made of sheets of fiberglass lying on top of one another and finished with a gel coating. An entrance **16** is formed within the housing **12** and is closed off or accessed by a door **14** positioned at the entrance. The door **14** also comprises a fiberglass material and is hingeably attached to the housing **12** at the entrance **16** such that it is openable and closable. The door **14** will include a seal (not shown) to ensure that water added to the interior of the housing **12** does not leak or otherwise disperse through the entrance **16** when the door is closed, such as shown in FIG. 1. Further aspects of the housing **12** include an interior wall **20** and an exterior wall **22**, and a bathtub top surface **28** extending generally between the interior **20** and exterior **22** walls. As shown in the figure, access panels **24**, **26** may be included on the external or internal walls to provide access to the interior of the housing walls such that pipes, electronics, or other internal components of the bathtub **10** can be

accessed. Still other components such as jets, control panels, user interfaces, faucets, showerheads, curtains, and the like may be included with the bathing environments.

(54) FIG. 2 shows a generally top planar view of a bathtub as similar to that shown in FIG. 1. The bathtub **10** shown in FIG. 2 includes many of the same components thereof, and includes a view of a top surface **28** generally connecting the interior wall **20** and exterior wall **22** of the housing **12**. Additional components shown in FIG. 2 include a seat **18**, a drain **32**, and a floor **34** of the housing **12**. The floor and/or seat may include a non-slip finish such that the finish will mitigate slippage by a user when the bathtub includes water added therein. Additional aspects may include a seat cushion on or around the seat **18** to provide comfort for the user of the bathtub **10**. Furthermore, as is shown in FIGS. 1 and 2, a grab bar **30** is included at the interior wall **20** of the housing **12**. The grab bar **30** shown in the figures is a substantially U-shaped member comprising a rigid material, such as metal, plastic, ceramic, or the like. In FIG. 1 the grab bar **30** is placed in a storage configuration, so as to produce an effect of being substantially flush with the interior wall **20**. In FIG. 2, the grab bar **30** is in a use configuration and as such extends outwardly from the interior wall **20**. As shown in FIG. 2 and additional figures that follow, there is an arrow pointing in an inward direction, wherein the storage configuration for the grab bar **30** will be such that a grabbing portion of the grab bar **30** is positioned inward whereas the use configuration (as of that shown in FIG. 2) is positioned opposite of the inward arrow.

(55) The grab bar **30** is angled on the interior wall of the housing **12**, to aid in stability for a user. For example, the grab bar **30** may be ergonomically angled to provide the user of the bathtub **10** in ease of sitting or standing with the least amount of effort. Furthermore, while it is not shown, additional grab bars can be positioned throughout the interior and/or exterior of the housing **12**, such as in or on the interior wall **20**, or the bars can be added as well as to the top surface **28** of the bathtub or shower or other bathing environment. Additional grab bars can provide additional stability when entering or exiting the walk-in type bathtub.

(56) FIGS. 3A-3B show a grab bar assembly **30**. The grab bar assembly **30** may include a handle **50**, a button **60**, a spring pin **70**, a central pivot pin **75**, a bolt **80** (shown in later figures), and a housing **90**.

(57) The grab bar assembly **30** is configured such that the handle **50** may be grabbed for stability when it is in a use configuration (as shown in FIG. 2). The grab bar **30** may alternatively be placed in a storage configuration (as shown in FIG. 1 and FIGS. 3A-3B). The handle **50** is rotatably attached to the housing **90** which houses the button **60**, the spring pin **70**, the central pivot pin **75**, and the bolt **80**. As such, the handle **50** is in rotatable communication with the housing **90**. The handle **50** is locked when in either of the use configuration or the storage configuration and can be unlocked by pressing the button **60** inward. By pressing the button **60** which is positioned on a viewable side of the housing **90**, the button **60** will interact with both of the spring pin **70** and the bolt **80**. When this happens, the handle **50** becomes unlocked, and the handle **50** may rotate around the central pivot pin **75** so as to transition between the use configuration and the storage configuration. The central pivot pin **75** is positioned transverse to the housing so as to allow the rotating handle **50** to be orthogonal or perpendicular to the housing **90** when in the use configuration, given that the central pivot pin **75** is a center point of rotation for the handle **50**.

(58) FIGS. 4-22B will elaborate on details of the handle **50**, the button **60**, the spring pin **70**, the central pivot pin **75**, the bolt **80**, and the housing **90**, and the remaining figures will return to explaining the grab bar **30** in greater detail after each of these components are explained. Further, FIGS. 3A, 3B, 4, 5, 6A, 6B, 7A, 7B, 19A, 19B, 20, 21A, 22B, 22A, 22B, 23A-C, 24A, 24B, 25A, and 25B are all scaled 1:2 in inches for the grab bar **30**. Further, FIGS. 8-17B are all scaled 3:1 in inches for the grab bar **30**.

(59) FIG. 4 shows a perspective view of the handle **50**. The handle **50** includes a handle grabbing portion **52**, a handle base **54**, a central pivot pin aperture **55**, a catch tab **56**, a first lock hole **58a**, and a second lock hole **58b**. FIGS. 5, 6A, 6B, 7A, and 7B show further views of the handle **50**.

Note that a back view of FIG. 5 is a mirror image of the front view shown in FIG. 5, whereas FIGS. 6A, 6B, 7A, and 7B when flipped 180° do not have a mirror image of one another and as such an “A” and a “B” figure are shown for each of these views.

(60) The handle **50** is configured such that the grabbing portion **52** extends outwardly from the handle base **54**. The grabbing portion **52** can be generally a U-shape, or otherwise any shape suitable for grabbing. Thus, the grabbing portion of **52** can be ergonomic for a user to grab and stabilize themselves with. As such, the grabbing portion **52** may extend away from the handle base **54** such that there includes a separation or opening between the handle base **54** and the grabbing portion **52**.

(61) The handle base **54** may include the central pivot pin aperture **55** such that the central pivot pin aperture **55** is centered within the handle base **54**. However, the central pivot pin aperture **55** will not be in a center of the handle base **54** such that it extends in a same direction as the grabbing portion **52**, rather the aperture for central pivot pin **55** will extend through the handle base **54** in a direction orthogonal to a direction which the grabbing portion **52** extends from the handle base **54**. This will allow for the handle **50** to rotate around the central pivot pin aperture **55** when the handle **50** is positioned within the grab bar assembly **30**.

(62) The handle **50** further includes the catch tab **56** to assist with handle **50** operation safety. As will be shown hereafter in later figures the catch tab **56** can make it so the handle **50** only rotates 180° and thus reduce a possibility of a user's hands entering the grab bar assembly **30** in an unexpected or dangerous manner. This does not mean that the catch tab **56** is a necessary component. Rather, the catch tab **56** can improve usability of the grab bar **30**. The handle **50** further includes the first lock hole **58a** and the second lock hole **58b** positioned such that when the handle **50** rotates around the central pivot pin aperture **55**, a bolting mechanism will be able to latch into either of the first lock hole **58a** or the second lock hole **58b** depending on whether the grab bar **30** is in the storage configuration or the use configuration.

(63) FIG. 8 shows a perspective view of the button **60**. The button **60** includes a button top **62**, a button bottom **64**, a pair of button columns **66**, and a pair of button slides **68**. The pair of button columns **66** and the pair of button slides **68** may exist as a single column and/or a single slide rather than each being in a pair. FIGS. 9A, 9B, 10A, 10B, 11A, and 11B show additional views of the button **60**.

(64) Generally, the button top **62** is positioned above the pair of button columns **66** which is positioned above the button bottom **64**. Further, the pair of button slides **68** generally exist on, and extend outwardly from, the pair of button columns **66**. As shown, the button **60** has generally a square shape for the button top **62**, however, other user-friendly shapes are possible. The button top **62** is intentionally wider than the button bottom **64** so that an operator who presses the button **60** may be able to access the button **60** with ease. The pair of button slides **68** include a generally sloped or angled shape so as to interact with another component of the grab bar assembly **30** smoothly. This will be described in detail when those figures are discussed. The button bottom **64** can serve as a connection between the pair of columns **66** when the pair of columns **66** exist as a pair rather than existing as a single column **66**. Should there exist a single column **66** rather than a pair of button columns **66**, the bolt **80** shown hereafter would need to be modified so as to go around the single button column **66** rather than through the pair of button columns **66**.

(65) FIGS. 12 and 13 show the spring pin **70** and the central pivot pin **75**. The spring pin **70** will work in conjunction with the bolt **80** which will be shown hereafter. The central pivot pin **75** need not be a screw, but preferably can be a screw. The central pivot pin **75** need only be a component that functions as a fulcrum for the handle **50** to rotate around, while also maintaining the handle **50** in between a recessed portion of the housing **90**.

(66) FIG. 14 shows a perspective view of the bolt **80**. The bolt **80** includes a bolt head **82**, a bolt tail **84**, a bolt aperture **85**, and a pair of bolt slides **88**. FIGS. 15A, 15B, 16A, 16B, 17A, and 17B show additional views of the bolt **80**.

(67) The bolt **80** is shaped so as to lock within the handle **50** via the bolt tail **84**. It is preferable that the bolt tail **84** is thick enough to withstand forces that will be acted on the grab bar by a user such that the bolt **80** will hold the handle **50** in place. The bolt head **82** includes the pair of bolt slides **88** which will interact with the pair of button slides **68** of the button **60**. As such, the pair of bolt slides **88** are shaped to interact with the pair of button slides **68**, and given that the pair of button slides **68** are generally sloped or angled in shape, it is preferred that the pair of bolt slides **88** are also generally sloped or angled in shape to match the angle thereof. A shape of both the button slide **68** and the bolt slides **88** need not be sloped or angled, rather the shape of the button slide **68** and the bolt slide **88** need only be able to interact with one another such that translational movement of the button **60** in a first direction will result in translational movement of the bolt **80** in a second direction orthogonal to the first direction that need button **60** was moved in.

(68) FIGS. **18A-18B** show perspective views of the housing **90**. The housing **90** includes a central pivot pin aperture **92**, mounting apertures **94**, a flange **96**, a viewing side **97**, a hidden side **98**, a recessed portion **100**, a handle aperture **102**, a button aperture **104**, a fitted button aperture **105**, a spring pin aperture **106**, a bolt aperture **108**, a first catch indent **110a**, a second catch indent **110b**, and a canyon **112**. FIGS. **19A**, **19B**, **20**, **20A**, **21B**, **22A**, and **22B** show additional views of the housing **90**. Further, FIG. **22A** acts as a reference perspective view for the cross-sectional view shown in FIG. **22B**.

(69) The housing **90** is designed to house a rest of the components of the grab bar **30** such that each of the rest of the components interact with the housing **90**, or are supported by the housing **90**. To aid in housing the components, the housing **90** includes a number of apertures to assist in supporting the components and easing their interacting with one another.

(70) The housing **90** includes the central pivot pin aperture **92** within the recessed portion **100** so as to receive the central pivot pin **75** and act as a fulcrum point for the handle **50** to rotate around. The central pivot pin aperture **92** is intentionally placed at a midpoint between the first catch indent **110a** and the second catch indent **110b**. This is so that when the handle **50** is rotated it will allow for the catch tab **56** to secure itself in either of the first catch indent **110a** or the second catch indent **110b** to assist the handle **50** in transitioning from the use configuration and the storage configuration.

(71) The housing **90** includes mounting apertures **94** at either end of the housing **90**. This is not limiting as to where the mounting apertures **94** are placed, rather, placing them on either end of the housing **90** assists in stability of the housing **90** and ultimately the grab bar **30**. The housing **90** may include more than two mounting apertures **94**. It is further contemplated that the mounting apertures **94** not be apertures at all, but could also be connection points to connect to a surface via known connecting and/or fastening methods not limited to but given by way of example glues or other adhesives, welding spots, lock-fitting tabs, etc.

(72) The housing **90** further includes the flange **96** extending outwardly which includes the viewing side **97** and the hidden side **98**, wherein the viewing side **97** is a side of the housing **90** on which the button **60** will be visible for pressing/operating. When looking down at the viewing side **97**, the housing **90** may be 10-20 inches in length and 1-4 inches in width. The viewing side **97** is designed to be aesthetically pleasing while also serving utility purposes such as being substantially flush with a surrounding surface (that the grab bar assembly **30** is mounted to) when the handle **50** is in the storage configuration (thus reducing obstacles that someone could contact should they inadvertently fall), and providing as much grabbing room as possible for the handle **50** when the handle **50** is in the use configuration. The hidden side **98** on the other hand is not viewable when the housing **90** is installed and will include the recessed portion **100** on a same side as the hidden side **98**. The recessed portion **100** may house all components except for components used for mounting. This reduces an amount of room or space needed to be removed from a surface which the grab bar **30** may be mounted to by keeping the mounting apertures **94** separate from, rather than included in, the recessed portion **100**.

(73) For example, the recessed portion **100** includes the handle aperture **102**. The handle aperture **102** is configured to house the handle **50** while allowing room enough for the handle **50** to rotate. The handle aperture **102** may be substantially an oval or stadium shape, or any other shape suitable to a shape of the handle **50**. In the figures, the recessed portion **100** only extends so far as to provide a little clearance for both the central pivot pin aperture **92** and the spring pin aperture **106** to not be right at an edge of the recessed portion **100**. However, the recessed portion **100** may extend further so as to entirely encompass or surround the handle **50** when the handle **50** rotates into the storage configuration. Benefits exist as to both scenarios. For example, extending the recessed portion **100** so as to fully surround the handle **50** when the handle **50** is rotated into the storage configuration can provide greater protection and longevity of the handle **50**, whereas keeping the recessed portion **100** shorter as shown in the figures reduces material costs. As shown in the figures, the handle aperture **102** extends entirely through the housing **90**. The handle aperture **102** thus extends through the central pivot pin apertures **92** so as to place the handle **40** therebetween and allow for lateral support of the handle **50** from both sides of the recessed portion **100** with the central pivot pin **75** acting as a load-bearing crossbar supported on either end by the recessed portion **100** with the handle **50** couched therebetween in the housing aperture **102**.

(74) The housing **90** further includes the button aperture **104** at a first side of the housing **90** to receive the button **60** and further includes the fitted button aperture **105** so as to guide the button bottom **64** and stabilize the button **60** when pressed. The fitted button aperture **105** may thus assist the button **60** by giving the button **60** stability if the fitted button aperture **105** is shaped the same as a silhouette of, or an exterior shape of, the button **60**. Further, the button aperture **104** is conveniently placed near the handle aperture **102** such that the user or operator may conveniently rotate the handle **50** right next to the button **60** when the button **60** is pressed.

(75) Included at a first end of the button aperture **104** is the spring pin aperture **106**. The spring pin aperture **106** is sized to surround the spring pin **70**. The spring pin aperture **106** may extend through the recessed portion **100** for ease in assembly of the grab bar **30**, and for stabilizing (keeping straight) the spring pin **70** as the spring pin **70** will be held in a lateral direction by the spring pin aperture **106**, and held in a longitudinal direction between the bolt **80** and a wall which the mounting apertures **94** provide a way of connecting to. The spring pin **70** may assist in biasing the bolt **80** towards the handle **50**. As such, included at a second end of the button aperture **104**, or an end opposite the first end of the button aperture **104** is the bolt aperture **108**. Similar to the spring pin aperture **106**, the bolt aperture **108** is sized to allow for its respective component to slide therethrough. The bolt aperture **108** therefore is generally or substantially a same shape as the bolt tail **84** so as to receive the bolt **80** and direct it to the handle **50** where the bolt **80** will be able to engage and lock with the handle **50** when the handle **50** is rotated into either of the use configuration or the storage configuration.

(76) As mentioned earlier, the first catch indent **110a** and the second catch indent **110b** exist on opposite sides of the housing **90**. Included with the first catch indent **110a** is the canyon **112**. The canyon **112** allows for the catch tab **56** to fully rotate from the use configuration to the storage configuration (and vice versa) without getting caught on the recessed portion **100**.

(77) FIGS. 23A, 23B, 23C, 24A, 24B, 25A, 25B, and 26A-C will be discussed together as they each represent different views of the grab bar **30** with all components assembled together. In these figures, the grab bar **30** is in a storage configuration similar to that shown in FIG. 1.

(78) FIGS. 23A and 23B show opposite sides of the grab bar assembly **30**. Note how the handle **50** appears off-center in these figures in relation to the mounting apertures on either end of the grab bar **30**. However, the handle **50** only appears so because of the button **60** is positioned to a side of the handle **50**, and the handle **50** is centered around the central pivot pin **75**. Thus, when installing the grab bar assembly **30** it is important to remember to install it such that the handle **50** will be accessible to the user without having to reach an uncomfortable distance. It may be preferable to have the button **60** closer to where a user sits, while on the other hand it may be preferable to have

the button **60** farther away such that the button **60** is accessed upon entering the bathtub but the handle **50** remains close to the user so as to adjust a seating position while the user is sitting in the bathtub.

(79) FIG. **23C** shows a mid-center cross-sectional view of the grab bar assembly **30** from the perspective of FIG. **23B**. As such, the interaction between the bolt tail **84** and the first lock hole **58a** can be seen. When the button top **62** is pressed inward it will cause the bolt tail **84** to remove itself from the first lock hole **58a**, and the handle **50** will then be freely rotatable. Notice that the bolt **80** is not aligned with the central pivot pin **75**. As such, this requires that the first lock hole **58a** and the second lock hole **58b** be placed equidistant from the central pivot pin **75** in opposite directions. Thus, when the handle **50** rotates into the use configuration, the second lock hole **58b** will be in alignment with the bolt tail **84** and ready to receive the bolt **80**. When the second lock hole **58b** is on a same side as the second catch indent **110b** (and thus the first lock hole **58a** would be on a same side as the first catch indent **110a**), then the grab bar **30** is in the use configuration.

(80) Note further how distal ends of the handle grab portion **54** are flush with the handle aperture **102** of the recessed portion **100**. In this regard, it is helpful to have the canyon **112** so that the catch tab **56** will be allowed to rotate an entire 180-degrees.

(81) FIG. **24A** provides additional perspective on these elements just discussed and how they interact with another. Namely, note the positioning of the handle grab portion **52** in relation to the first catch portion **110a**. Note how the catch tab **56** would rotate directly into that first catch portion **110a** should the handle **50** rotate around the central pivot pin **75**. Note further how the bolt slides **88** are visible and placed directly on the button slides **68** and how the bolt tail **84** is visible after passing through the recessed portion **100**. The spring pin **70** is assisting in biasing the bolt **80** into the handle **50** when the spring pin **70** is compressed against a wall. Note even further how the button bottom **64** is visible in this figure. This is because the bolt **80** extends through the button **60** and can thus more easily assist in translating a first motion into a second motion orthogonal to the first motion. Shaping the housing **90** and the button **60** such that the bolt **80** extends through the button is a superior design not only for its ease in activating the lock/unlock feature, but also for the bolt's **80** improved stability being surrounded by both the button **60** and a portion of the recessed portion **100**.

(82) FIG. **24B** shows a side of the grab bar **30** which will be visible to the user after installation. The button top **62** provides ease of access to lock and unlock the grab bar **30**.

(83) Finally, FIGS. **26A-26C** provide a zoomed-in view of the locking mechanism. FIG. **26A** provides a reference figure for both FIG. **26B** and FIG. **26C** in that the latter are both cross-sectional views of FIG. **26A** at varying depths into the grab bar **30**.

(84) As shown in these figures, the button top **62** has some room to be pressed inward. When the button **60** is pressed inward, this will force an interaction between the pair of button slides **68** and the pair of bolt slides **88**. As the button **60** travels further inward, the bolt head **82** will slide further and further away from the handle **50**, thus removing the bolt tail **84** from the first lock hole **58a** (or the second lock hole **58b** if the handle **50** is in the use configuration). When the button top **62** is pressed as far inward as it will go, this will place the spring pin **70** at a maximum potential energy so as to urge the pair of bolt slides **88** back along the pair of button slides **68** and towards the handle **50** if a user removes force from pressing the button **60** inward, wherein the button top **62** will be urged back a direction opposite the inward arrow shown from the motion of the bolt **80** towards the handle **50** via the spring pin **70**.

(85) It is to be further appreciated that the terms of the application are not to be limiting and all-inclusive. For example, while the terms use and non-use have been used in relation to the grab bars, it is to be appreciated that the bars could be used in their collapsed manner, such as by attaching an anchor, rope, or other device thereto to aid in use of the tether or other device. The bars need not be used in a bathing environment or any other specific environment for that matter.

(86) For example, the grab bar **30** may particularly be associated with any type of seat, including

stadium seating, theater seating, boating, automobile seating, airline seating, outdoor seating, classroom seating, hospital seating, or the like. The bars may also act as concealable or retractable handles for movable objects and may be associated with reusable packaging, other moving materials, storage containers, luggage, laundry baskets, or the like.

(87) According to another nonlimiting example, the bars may also act as concealable or retractable handles for substantially movable objects and may be associated with kitchen appliances, grills, doors, or the like.

(88) According to another nonlimiting example, the bars can be used in the fitness industry to facilitate or resist certain types of exercises when mounted on a wall. The grab bars can flip over and be used with bands for stretching. The grab bars may be installed in the floor so that the grab bars may be gripped while a user does push-ups or other types of exercises where you are elevated to work your triceps or biceps.

(89) According to another nonlimiting example, the grab bars can be used in closets at home or can be used in an office setting. The bar of the grab bars may be substituted for hooks and installed in a wall or other surface and when needed can be flipped into the use configuration so that clothes may be hung. When no longer needed, the hooks may be returned to the non-use configuration such that they are concealed and a flush surface is created. The hooks or grab bars may be installed within a concrete, outdoor wall, or other hard surface to create a place to tie a dog leash such that a dog does not have the ability to run away or attack persons passing by.

(90) The bars could also be maintained in their extended configuration even when not in use. Furthermore, it is to be appreciated that additional holding and moving mechanisms could be used to move the bars between a generally extended and retracted or housed position.

(91) From the foregoing, it can be seen that the present invention accomplishes at least all of the stated objectives.

LIST OF REFERENCE CHARACTERS

(92) The following table of reference characters and descriptors are not exhaustive, nor limiting, and include reasonable equivalents. If possible, elements identified by a reference character below and/or those elements which are near ubiquitous within the art can replace or supplement any element identified by another reference character.

TABLE 1: LIST OF REFERENCE CHARACTERS

(93)	10	walk-in-bathtub	12	housing	14	door	16	entrance	18	seat	20	interior wall	22	exterior wall	24	access panel	26	access panel	28	top surface	30	grab bar	32	drain	34	floor	50	handle	52	handle grabbing portion	54	handle base	55	central pivot pin aperture	56	catch tab	58a	first lock hole	58b	second lock hole	60	button	62	button top	64	button bottom	66	button column	68	button slide	70	spring pin	75	central pivot pin	80	bolt	82	bolt head	84	bolt tail	85	bolt aperture	88	bolt slide	90	housing	92	central pivot pin aperture	94	mounting apertures	96	flange	97	viewing side	98	hidden side	100	recessed portion	102	handle aperture	104	button aperture	105	fitted button aperture	106	spring pin aperture	108	bolt aperture	110a	first catch indent	110b	second catch indent	112	canyon
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Glossary

(94) Unless defined otherwise, all technical and scientific terms used above have the same meaning as commonly understood by one of ordinary skill in the art to which embodiments of the present disclosure pertain.

(95) The terms “a,” “an,” and “the” include both singular and plural referents.

(96) The term “or” is synonymous with “and/or” and means any one member or combination of members of a particular list.

(97) As used herein, the term “exemplary” refers to an example, an instance, or an illustration, and does not indicate a most preferred embodiment unless otherwise stated.

(98) The term “about” as used herein refers to slight variations in numerical quantities with respect to any quantifiable variable. Inadvertent error can occur, for example, through use of typical measuring techniques or equipment or from differences in the manufacture, source, or purity of

components.

(99) The term “substantially” refers to a great or significant extent. “Substantially” can thus refer to a plurality, majority, and/or a supermajority of said quantifiable variables, given proper context.

(100) The term “generally” encompasses both “about” and “substantially.”

(101) The term “configured” describes structure capable of performing a task or adopting a particular configuration. The term “configured” can be used interchangeably with other similar phrases, such as constructed, arranged, adapted, manufactured, and the like.

(102) Terms characterizing sequential order, a position, and/or an orientation are not limiting and are only referenced according to the views presented.

(103) The “invention” is not intended to refer to any single embodiment of the particular invention but encompass all possible embodiments as described in the specification and the claims. The “scope” of the present disclosure is defined by the appended claims, along with the full scope of equivalents to which such claims are entitled. The scope of the disclosure is further qualified as including any possible modification to any of the aspects and/or embodiments disclosed herein which would result in other embodiments, combinations, subcombinations, or the like that would be obvious to those skilled in the art.

Claims

1. A grab bar assembly, comprising: a housing including a recessed portion and a flange portion; a central pivot pin positioned in the recessed portion; a handle comprising a grabbing portion, a handle base, a first lock hole, and a second lock hole, wherein the handle is rotatable around the central pivot pin between a use configuration and a storage configuration; a button on the flange portion extending into the recessed portion and positioned on a first side of the handle, the button including a pair of button slides; and a bolt extending through the button, through the recessed portion, and into either of the first lock hole or the second lock hole; wherein the bolt includes a pair of bolt slides in sliding communication with the pair of button slides so as to be capable of converting a first translational motion into a second translational motion orthogonal to the first translational motion when the pair of button slides interface with the pair of bolt slides.
2. The grab bar assembly of claim 1, wherein the central pivot pin passes through the recessed portion, the handle base, and the recessed portion again.
3. The grab bar assembly of claim 1, wherein an end of the central pivot pin is male-threaded and a central pivot pin aperture receiving the central pivot pin is female-threaded.
4. The grab bar assembly of claim 1, further comprising a spring pin to bias the bolt towards the handle, wherein the spring pin extends through the recessed portion and into the bolt.
5. The grab bar assembly of claim 1, wherein the handle includes a catch tab for preventing the rotatable body support from rotating more than 180°.
6. The grab bar assembly of claim 5, wherein the housing includes a first catch indent positioned at a second side of the handle, wherein the housing includes a second catch indent positioned at the first side of the handle, and wherein when the handle rotates from the use configuration to the storage configuration a first side of the catch tab interfaces with the first catch indent and a second side of the catch tab interfaces with the second catch indent.
7. The grab bar assembly of claim 6, further comprising a canyon in the recessed portion to allow for a full 180° range of motion for the handle.
8. The grab bar assembly of claim 1, wherein the button further includes a button top substantially flush with the flange, a button bottom opposite the button top, and a pair of button columns connecting the button bottom and the button top.
9. The grab bar assembly of claim 8, further comprising a fitted button aperture to receive and stabilize the button bottom.
10. The grab bar assembly of claim 8, wherein the bolt includes a bolt head and a bolt tail opposite

the bolt head for inserting into the handle, wherein the bolt head is wider than the bolt tail.

11. The grab bar assembly of claim 10, wherein the bolt tail extends through the pair of button columns, the button top, and the button bottom, but the bolt head cannot extend through the pair of button columns, the button top, and the button bottom.
 12. The grab bar assembly of claim 11, wherein the button bottom prevents the button top from extending further than the flange.
 13. The grab bar assembly of claim 12, wherein the pair of button slides extend from the pair of button columns.
 14. The grab bar assembly of claim 1, wherein the first lock hole and the second lock hole are not aligned across the handle base.
 15. The grab bar assembly of claim 1, further comprising mounting apertures positioned at distal ends of the flange.
 16. The grab bar assembly of claim 1, wherein the spring pin extends through a button aperture of the grab bar assembly.
 17. The grab bar assembly of claim 1, wherein the handle base is substantially flush with the flange when in the storage position.
 18. The grab bar assembly of claim 1, wherein the housing provides stability for each of the handle, the button, the bolt, the spring pin, and the central pivot pin.
 19. A grab bar assembly built into a surface comprising: a grab bar housing including a recessed portion and a flange portion; a central pivot pin positioned in the recessed portion; a handle comprising a grabbing portion, a handle base, a first lock hole, and a second lock hole, wherein the handle is rotatable around the central pivot pin between a use configuration and a storage configuration; a button on the flange portion extending into the recessed portion and positioned on a first side of the handle, the button including a button slide; and a bolt extending into either of the first lock hole or the second lock hole, the bolt including a bolt slide for interfacing with the button slide; wherein in the storage configuration the grabbing portion is concealed within a wall of a structure, said structure being a building, a seat, or an appliance; wherein the grabbing portion is at or below the surface when in the storage configuration; and wherein the grabbing portion extends outward at least partially from the surface in the use configuration.
 20. A method of concealing and revealing a handle, comprising: housing the handle comprising a grabbing portion, a handle base, and a central pivot pin aperture with a central pivot pin extending therethrough within a housing having a recessed portion and a flange, said recessed portion both receiving and supporting the central pivot pin and said flange extending outwardly of the recessed portion and having a thickness less than a depth of the recessed portion; allowing the rotatable body support to rotate approximately 180° around the central pivot pin between a use configuration and a storage configuration, wherein in the storage configuration the housing conceals the grabbing portion within a surface of a building, a seat, or an appliance; and pressing a button positioned on the flange to lock or unlock the handle to allow for rotation from the use configuration to the storage configuration; wherein the grab bar housing is mounted to the building, the seat, or the appliance such that at least a portion of the grab bar housing is flush with the surface.
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